

H2 Chemistry 9647**2013 AJC2 Prelim P3 Suggested Solutions**

- 1 (a) (i) Since $n = 2$, [1]

$$E = E(\text{pH } 7) - \frac{0.0592}{2} \log_{10} \frac{[\text{NADH}]}{[\text{NAD}^+]}$$

$$-0.350 = -0.320 - \frac{0.0592}{2} \log_{10} \frac{[\text{NADH}]}{[\text{NAD}^+]}$$

$$\log_{10} \frac{[\text{NADH}]}{[\text{NAD}^+]} = 1.01$$

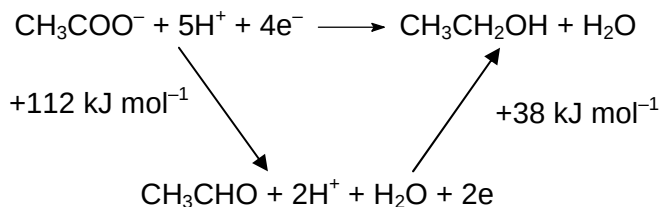
$$\frac{[\text{NADH}]}{[\text{NAD}^+]} = \mathbf{10.3} \quad [1]$$

(ii) % of NAD^+ = $\frac{1}{(10.3 + 1)} \times 100\%$ [1]
 = **8.85 %**

(iii) Stage 1: [2]
 $E_{\text{cell}}(\text{pH } 7) = -0.320 - (-0.197)$
 = **-0.123 V**

Stage 2:
 $E_{\text{cell}}(\text{pH } 7) = -0.320 - (-0.581)$
 = **+0.261 V**

[1] each stage

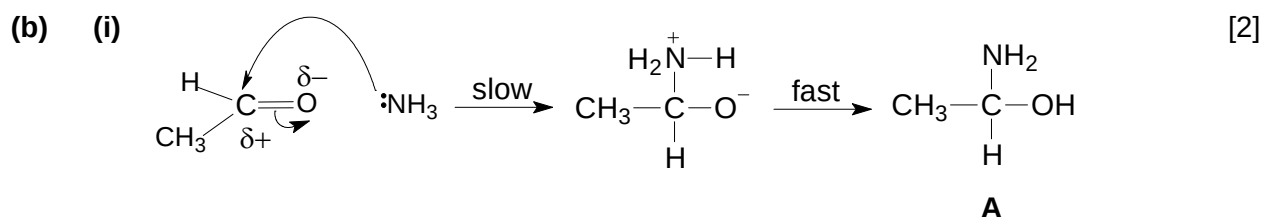


$$\Delta G(\text{pH } 7) = +112 + (+38)$$

$$= \mathbf{+150 \text{ kJ mol}^{-1}} \quad [1]$$

(v) $\Delta G(\text{pH } 7) = -zFE(\text{pH } 7)$ [2]
 $E(\text{pH } 7) = \frac{-150 \times 10^3}{(4)(96500)}$
 = **-0.389 V**

[1]: correct value for z
 [1]: correct $E(\text{pH } 7)$ calculated



[1] correct dipoles on the carbonyl C and O and curly arrows for slow step
 [1] correct intermediate

(ii) **elimination** [1]

(iii) **No.** [1]

The CH_3CHO molecule is trigonal planar with respect to the carbonyl carbon atom. Since the molecule is planar, the NH_3 nucleophile has equal probability of attacking it from either side of the molecule. [1]

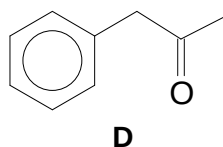
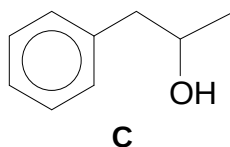
This generates equimolar mixture of the enantiomers (racemic mixture). Each enantiomer rotates plane polarised light in different directions by the same extent and their optical activity are cancelled out. [1]

(c) (i) **electrophilic substitution** [1]

(ii) Steps V: (cold) concentrated H_2SO_4 followed by H_2O , warm Or $\text{H}_2\text{O(g)}$, concentrated H_3PO_4 , 300°C , 70 atm [4]

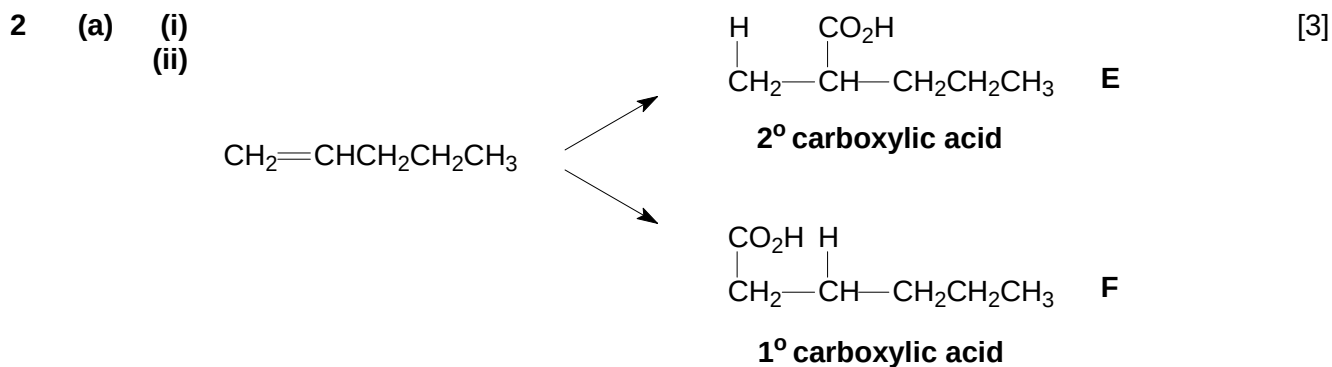
Steps VI: $\text{K}_2\text{Cr}_2\text{O}_7$, dilute H_2SO_4 , heat (under reflux)

(cannot accept KMnO_4 , dilute H_2SO_4 , heat for step VI)



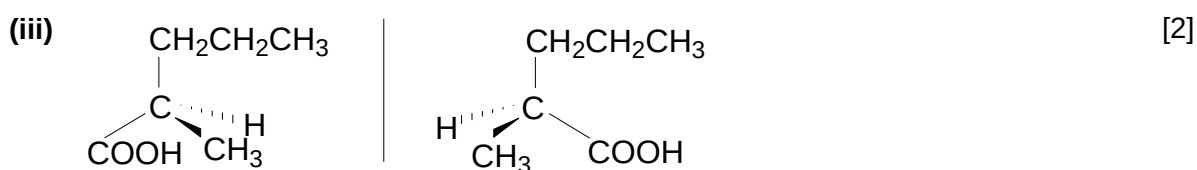
[1] each: correct structure for **C** and **D**

[1]: correct reagent and conditions for each stage



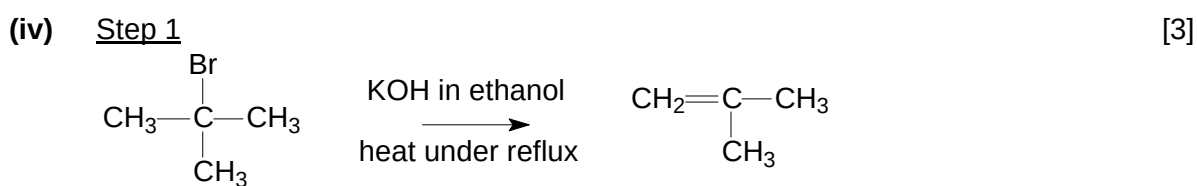
[1] each: isomer **E** and **F** correctly drawn and labelled

[1]: correct classification of both carboxylic acids



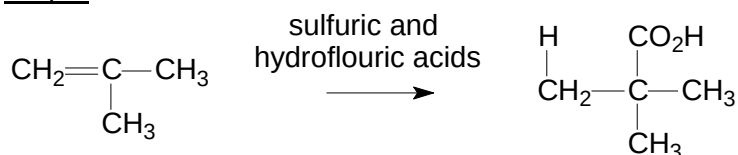
[1]: correct isomer (**E**) identified

[1]: correct mirror images shown



(accept $(\text{CH}_3)_2\text{CHCH}_2(\text{Br})$ as starting material)

Step 2



[1]: correct isomer of $\text{C}_4\text{H}_9\text{Br}$ identified

[1]: correct reagent and condition for Step 1

[1]: correct intermediate formed

(b) (i) To generate a more stable carbocation intermediate [1]

(ii) **Nucleophile** [1]

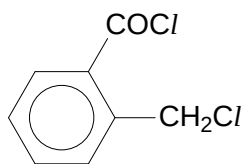


(iii) any isomer of hexene (because rearrangement to a 3° carbocation will occur) [1]

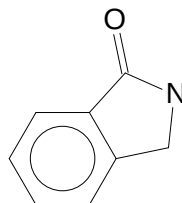
- (iv) The steric hindrance due to the 3 bulky R groups present in 3° carboxylic acid prevents the attack of the nucleophile on the carbonyl carbon during the hydrolysis reaction. [1]

(accept: electron-donating effect of alkyl groups making the carbonyl carbon less electron deficient and hence less susceptible to nucleophilic attack by water)

(c) (i)



J



K

[1] each

- (ii) (base-catalysed) **hydrolysis** [1]

- (iii) $\text{PCl}_5 + 4\text{H}_2\text{O} \longrightarrow \text{H}_3\text{PO}_4 + 5\text{HCl}$ [1]
 PCl_5 hydrolyse in water to produce toxic, acidic gas and not available to react with SO_2 . [1]

OR

$\text{SOCl}_2 + \text{H}_2\text{O} \longrightarrow \text{SO}_2 + 2\text{HCl}$
 SOCl_2 hydrolyse in water to produce toxic, acidic gases and is not formed.

- (iv) $\text{SO}_2 + \text{PCl}_5 \longrightarrow \text{SOCl}_2 + \text{POCl}_3$ (**L**) [2]

[1]: correct identity for **L**

[1]: correct balanced equation

- 3 (a) (i) The products formed by S_N1 mechanism are racemic mixtures and do not exhibit optical activity / do not rotate plane-polarised light, [1]
 whereas the product formed by S_N2 mechanism has an inversion of stereoisomerism, thus the product will exhibit optical activity / rotate plane-polarised light. [1]

- (ii) Let the % of reactants that take part in the S_N2 mechanism be x . [1]

$$\frac{x(13.5) + (100 - x)(0)}{100} = 5.5$$

$$x = 40.7\%$$

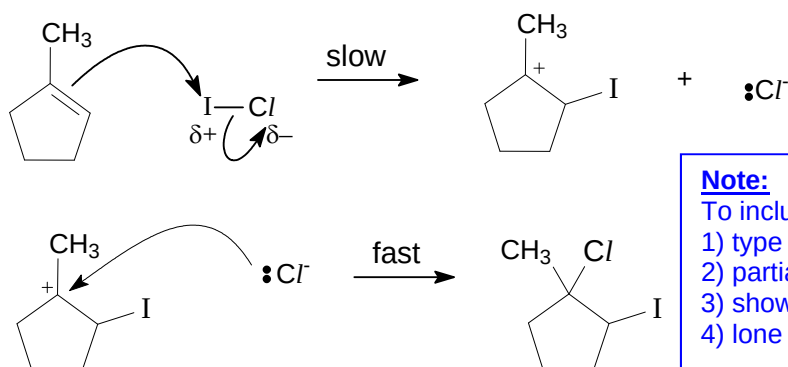
- (iii) The relative ease of hydrolysis of monohalogenoethanes, C_2H_5X can be determined by the bond energies of the $C-X$ bond. [1]

Since $C-Cl$ bond is stronger, 2-chlorobutane tends to hydrolyse slower than 2-iodobutane.

Note:

You may refer to the Data Booklet for bond energies of $C-I$ (240 kJ mol^{-1}) and $C-Cl$ (340 kJ mol^{-1}).

- (b) Type of reaction: **electrophilic addition** [3]



Note:

To include:

- 1) type of reaction,
- 2) partial charges on $I-Cl$ molecule,
- 3) show all curly arrows and
- 4) lone pairs on Cl^- ion.

[1]: type of reaction

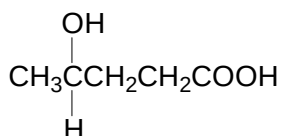
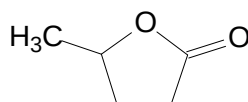
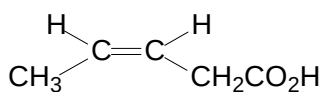
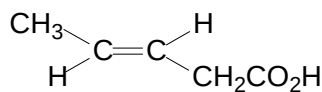
[1]: correct partial charges on $I-Cl$ and curly arrows, fast and slow steps

[1]: correct intermediate and product

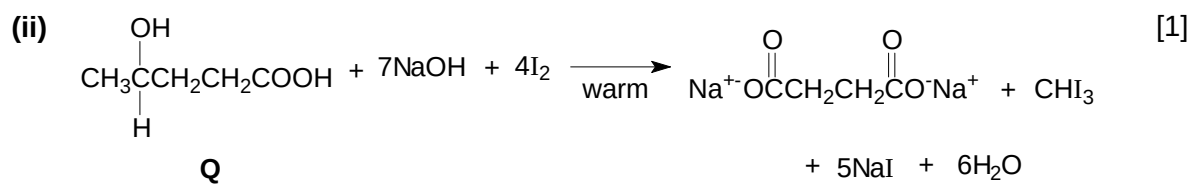
- (c) [2]

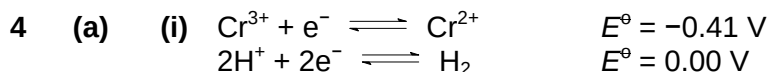
[1] each

(d) (i)	Observation	Type of reaction	Deduction
	P is sparingly soluble in water		(P is not a salt, but may contain hydroxyl or COOH group)
	P is readily soluble in NaOH	acid–base reaction	P contains a COOH group
	P reacts with hot aq NaOH to give Q , C ₅ H ₁₀ O ₃	nucleophilic substitution	Q contains an alcohol functional group.
	Q reacts with alkaline I ₂ (aq) to give yellow ppt.	oxidation (iodoform reaction)	Q contains CH ₃ CH(OH)– $\begin{array}{c} \text{CH}_3 \\ \\ -\text{C}=\text{O} \end{array}$ or $-\text{C}=\text{O}$ structure
	Q reacts with concentrated H ₂ SO ₄ to give R , which is a stable sweet-smelling liquid	self–esterification	R is an ester . (R must be a 5 or 6 membered cyclic ester as it is stable, hence minimal ring strain) Q contains both the –OH and –COOH group
	Q reacts with concentrated H ₂ SO ₄ to give S and T , with the formula C ₅ H ₈ O ₂	elimination	S and T contains C=C (S and T exhibits geometrical isomerism)
	S reacts with 1 mol of Br ₂ (aq)	electrophilic addition	S contains <u>only 1 C=C</u> (a bromohydrin product is formed as the major product)
	S and T reacts with KMnO ₄ to give U , C ₃ H ₄ O ₄ and ethanoic acid	(strong) oxidation (oxidative cleavage)	The C=C in S/T is cleaved, there is no terminal C=C as no CO ₂ is produced.
	1 mole of U reacts with excess Na ₂ CO ₃ to give 1 mole of CO ₂ .	acid–base reaction	$\begin{array}{c} \text{OH} \\ \\ -\text{C}=\text{O} \end{array}$ U contains 2 $-\text{C}(=\text{O})\text{OH}$ U is a <u>dicarboxylic acid</u> .

**Q****R****S****T****U**

[1] each: structures **Q**, **R**, **S**, **T** and **U**
 [5]: 14 – 18 pts for explanation
 [4]: 11 – 13 pts
 [3]: 8 – 10 pts
 [2]: 5 – 7 pts
 [1]: 1 – 4 pts





$$E_{\text{cell}}^\ominus = 0 - (-0.41) \\ = +0.41 \text{ V}$$

Since $E_{\text{cell}}^\ominus \geq 0$, reaction is feasible and the blue Cr^{2+} is oxidised to green Cr^{3+} . [1]



In presence of water ligands, the 3d-orbitals are split into 2 energy levels with a small energy gap that lies within the visible light spectrum.

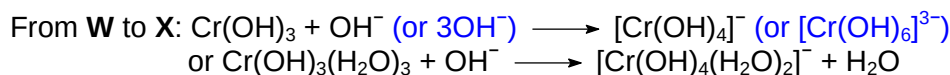
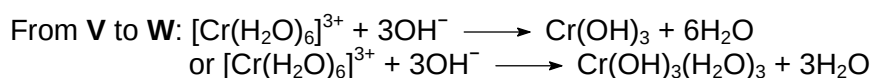
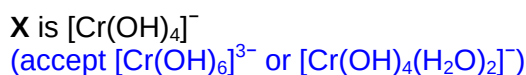
Upon absorption of light energy, the electron is excited from the lower energy 3d-orbitals to the vacant higher energy 3d-orbitals.

Color reflected, which is green, is complementary to the color absorbed, which is red.

[1]: partially filled 3d-orbital; (small) energy gap

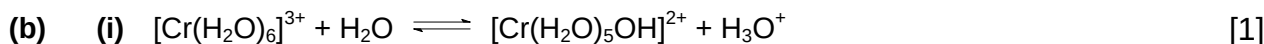
[1]: d-d transition

[1]: absorption of red (complementary to green)



[1] each: correct structure for **W** and **X**

[1]: both equations correctly balanced



$$K_a = \frac{[[\text{Cr}(\text{H}_2\text{O})_5\text{OH}]^{2+}][\text{H}_3\text{O}^+]}{[[\text{Cr}(\text{H}_2\text{O})_6]^{3+}]} \\ = \frac{[\text{H}_3\text{O}^+]^2}{[[\text{Cr}(\text{H}_2\text{O})_6]^{3+}]}$$

$$[[\text{Cr}(\text{H}_2\text{O})_6]^{3+}]_{\text{eqm}} = \frac{(10^{-3})^2}{9.8 \times 10^{-5}} \\ = 0.0102 \text{ mol dm}^{-3}$$

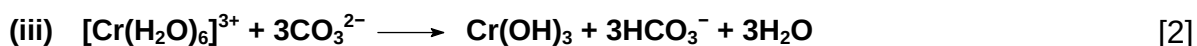
$$[[\text{Cr}(\text{H}_2\text{O})_6]^{3+}]_{\text{initial}} = [[\text{Cr}(\text{H}_2\text{O})_6]^{3+}]_{\text{eqm}} + [[\text{Cr}(\text{H}_2\text{O})_6]^{3+}]_{\text{ionised}} \\ = 0.0102 + (1 \times 10^{-3}) \\ = 0.0112 \text{ mol dm}^{-3}$$

(accept 0.0102 mol dm⁻³)



$$\frac{n(\text{CO}_3^{2-})}{n([\text{Cr}(\text{H}_2\text{O})_6]^{3+})} = \frac{0.000330}{0.000112} \approx \frac{3}{1}$$

(accept 3.2 – 3.3)



[1]: balanced ionic equation

[1]: correct by-product (HCO_3^-) formed

(c) (i) Denaturation of proteins involves the disruption of the secondary, tertiary and quaternary structures which are held together by relatively weak forces of attraction. [1]

(ii) Chromium destroys the disulfide bonds by reducing the S–S bonds in oxytocin to –SH groups. [1]

(iii) Energy from heating overcomes the van der Waal's forces between Gly and Leu (any pair from Gly, Leu, Pro, Ile) that hold the secondary, tertiary and quaternary structures together. [2]

or Heating overcomes the hydrogen bonding between Asn and Gln (any pair from Asn, Gln, Tyr)

[1]: correct interaction identified

[1]: correct illustration with suitable amino acids

(iv) dilute H_2SO_4 , heat for prolonged period (or aqueous NaOH) [1]



[1]: correct amino acid with displayed formula

[1]: both chiral carbons identified

- 5 (a) (i) $\text{P}_4\text{S}_3 + 8\text{O}_2 \longrightarrow \text{P}_4\text{O}_{10} + 3\text{SO}_2$ [2]
(P₄O₆ accepted although it is a liquid)
(ignore state symbol; do not accept P₂O₅, P₂O₃) [ecf for (ii) & (iii)]

[1]: correct oxide

[1]: balanced equation

- (ii) Both SO₂ and P₄O₁₀ have simple molecular structure with weak van der Waals forces between the molecules [1]
 SiO₂ has a giant molecular structure with strong, extensive covalent bonds between Si and O atoms [1]

hence SO₂ and P₄O₁₀ have m.p. that are similar but much lower than that of SiO₂.

- (iii) $\text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \longrightarrow 4\text{H}_3\text{PO}_4$ (pH < 2) [2]

[1]: balanced equation

[1]: correct pH

- (b) (i)
$$K_p = \frac{p_{\text{SO}_3}^2}{p_{\text{SO}_2}^2 p_{\text{O}_2}} \text{ atm}^{-1}$$
 [2]

[1]: correct expression

[1]: correct units

- (ii) Temperature should be low and pressure should be high. [1]

A low temperature will cause position of equilibrium will shift to the right to favour the forward exothermic reaction. Yield of SO₃ will increase. [1]

A high pressure will cause position of equilibrium will shift to the right to reduce the pressure as it results in a decrease in no. of gaseous particles. Yield of SO₃ will increase [1]

- (iii) Vanadium(V) oxide has partially-filled (vacant) 3d-orbitals available to form temporary bonds with the gaseous SO₂ and O₂ molecules. [1]

- (iv) For concentrated H₂SO₄ [2]
 $\text{I}^- + \text{H}_2\text{SO}_4 \longrightarrow \text{HI} + \text{HSO}_4^-$
 $8\text{HI} + \text{H}_2\text{SO}_4 \longrightarrow 4\text{I}_2 + \text{H}_2\text{S} + 4\text{H}_2\text{O}$

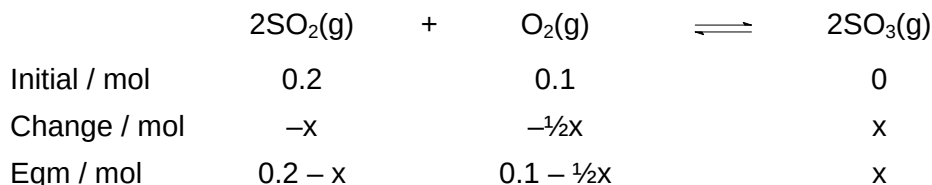
For concentrated H₃PO₄
 $\text{I}^- + \text{H}_3\text{PO}_4 \longrightarrow \text{HI} + \text{H}_2\text{PO}_4^{2-}$

[1] each acid

Note: conc. H₃PO₄ is not an oxidising agent like conc. H₂SO₄ hence will only reacts with NaI in an acid-base reaction.

(c) (i) $n(\text{NaOH}) = \frac{39.20}{1000} \times 1.00$ [1]
 $= 0.0392 \text{ mol}$
 $n(\text{H}^+) \text{ present in } 25 \text{ cm}^3 = 0.0392 \text{ mol}$
 $n(\text{H}^+) \text{ in } 250 \text{ cm}^3 \text{ standard flask} = \frac{250}{25.0} \times 0.0392$
 $= \mathbf{0.392 \text{ mol}}$

(ii) Let $n(\text{SO}_3)$ at equilibrium = $x \text{ mol}$



$$n(\text{H}_2\text{SO}_3) = (0.2 - x) \text{ mol}$$

$$n(\text{H}_2\text{SO}_4) = x \text{ mol}$$

$$\begin{aligned} n(\text{H}^+) &= n(\text{H}^+ \text{ from } \text{H}_2\text{SO}_3) + n(\text{H}^+ \text{ from } \text{H}_2\text{SO}_4) \\ &= (0.2 - x) + 2x \\ &= (0.2 + x) \text{ mol} \end{aligned} \quad [1]$$

$$0.2 + x = 0.392$$

$$x = 0.192$$

$$n(\text{SO}_3) \text{ at equilibrium} = \mathbf{0.192 \text{ mol}} \quad [1]$$

$$n(\text{SO}_2) = 0.2 - 0.192 = \mathbf{0.008 \text{ mol}} \quad [1]$$

$$n(\text{O}_2) = 0.1 - (\frac{1}{2} \times 0.192) = \mathbf{0.004 \text{ mol}}$$

(iii) Total amount of gases = $0.192 + 0.008 + 0.004$
 $= 0.204 \text{ mol}$

$$p_{\text{SO}_3} = \frac{0.192}{0.204} \times 5 = 4.71 \text{ atm}$$

$$p_{\text{SO}_2} = 0.196 \text{ atm}$$

$$p_{\text{O}_2} = 0.0980 \text{ atm} \quad [1]$$

$$\begin{aligned} K_p &= \frac{4.71^2}{0.196^2 \times 0.098} \\ &= \mathbf{5.89 \times 10^3 \text{ atm}^{-1}} \end{aligned} \quad [1]$$